

Building Event-Driven AI Agents for Proactive SAP Incident Management

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TURNING QUALIFYING SAP CLOUD ALM ALERTS INTO CONTEXTUAL, HUMAN-GOVERNED RESOLUTION WORKFLOWS

Monitor. Investigate. Recommend. Approve. Resolve. Validate.

A practical multi-agent architecture on SAP Business Technology Platform that helps operations teams move from manual alert investigation to evidence-based incident resolution.

SAP Cloud ALM

Operational alert source

AI agents

Investigation and guidance

Teams / ServiceNow

Collaboration and governance



Cloud ALM Detects Issues. Resolution Still Requires Manual Effort

THE PROBLEM

An alert is the beginning of an incident – not the end of the work.

Technical alert, limited context

Operators need context on affected systems and services.

Manual investigation

Teams search logs, runbooks, and past incidents.

High coordination effort

Unclear who should resolve the issue.

Knowledge is fragmented

Fixes remain in tickets, not reusable guidance.

THE VISION

From Reactive Monitoring to Proactive AI-Assisted Operations

The goal is not to replace operators. It is to give them a system that has already completed the initial investigation before they begin manual troubleshooting.

1

1. Qualifying Alert

SAP Cloud ALM creates or updates an operational alert.

2

2. Agents Invoke

Relevant agents begin triage, classification, and correlation.

3

3. Context Assembled

Evidence, runbooks, historical incidents, and likely causes are retrieved.

4

4. Human-Governed Action

Recommendation is reviewed; approved action is executed and validated.

ARCHITECTURE OVERVIEW

A Multi-Agent System for Proactive SAP Incident Management

Built on SAP BTP with AI-assisted investigation and human-governed resolution.

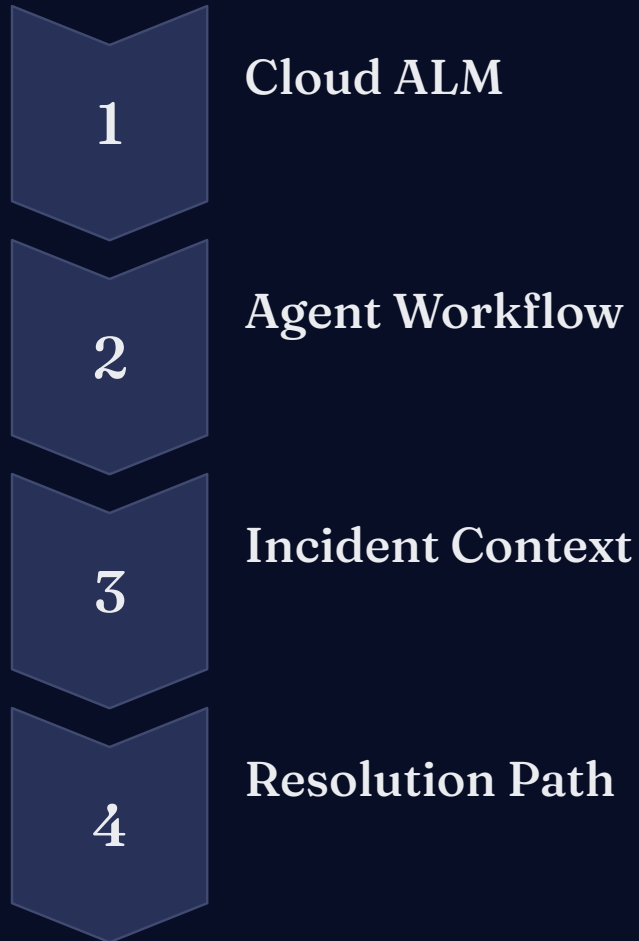
The architecture uses specialized agents with clear responsibilities. Rather than a monolithic AI component, agents investigate issues, retrieve knowledge, recommend next steps, and support controlled resolution. When SAP Cloud ALM creates or updates a qualifying alert, the relevant agents are invoked on demand.

- ❏ Architecture-first: grounded retrieval, human approval, and auditability are core design patterns.



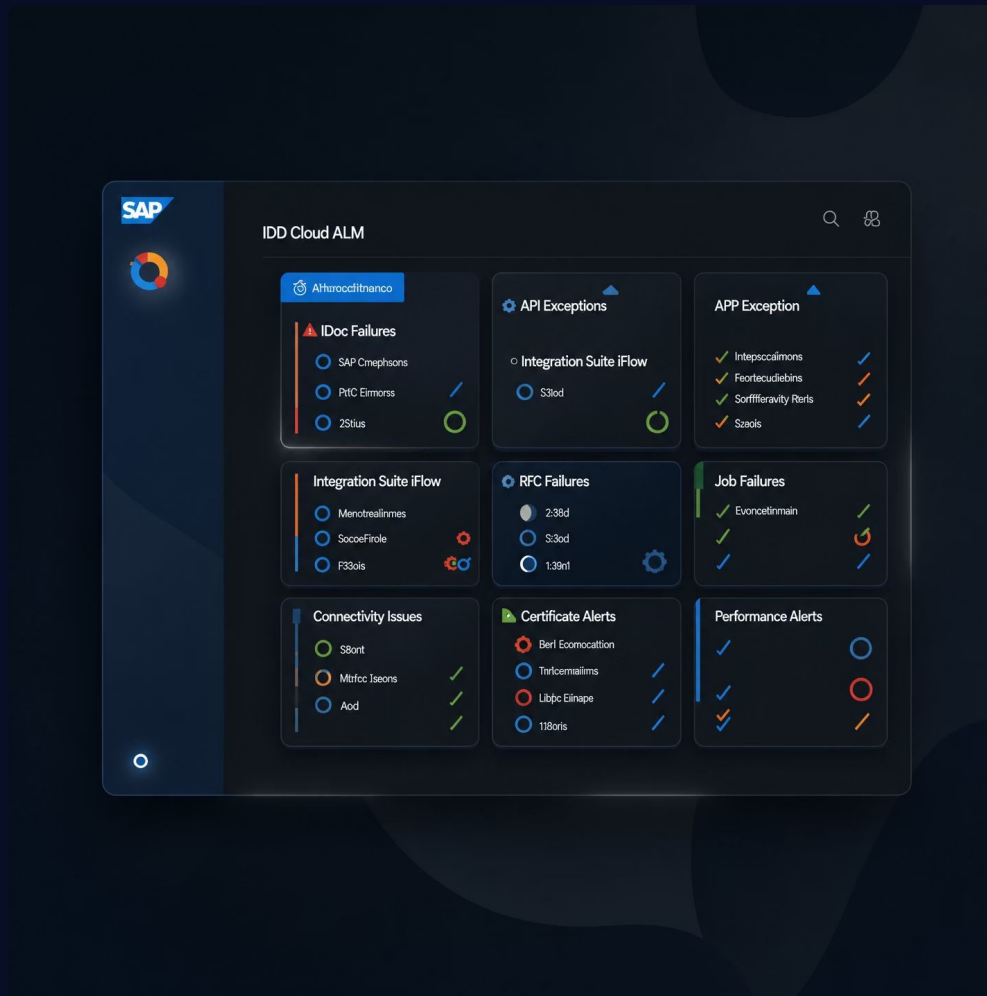
TRIGGER MODEL

Cloud ALM Alerts Invoke Agents on Demand



SCOPE

One Framework for Multiple Operational Alert Types



The architecture applies to qualifying operational alerts across the SAP landscape.

- Integration Suite message and iFlow failures
- IDoc, RFC, proxy, and API exceptions
- Application errors and technical exceptions
- Failed jobs and scheduled activities
- Connectivity, authentication, and authorization issues
- Health, availability, performance, and capacity alerts

CONCRETE EXAMPLE: IDoc Failure Walkthrough

1

1. Monitor

Cloud ALM detects: "IDoc ORDERS_05 failed to post to SAP ERP. Retry exhausted after 3 attempts."

2

2. Investigate

Agents retrieve: IDoc payload, ERP posting logs, recent GL account changes, known errors for ORDERS_05, prior similar incidents from 6 months ago.

3

3. Recommend

Agent proposes: "GL account 4100 was locked yesterday during month-end close. Unlock account and retry IDoc. Estimated resolution time: 5 minutes."

4

4. Approve

Operator reviews recommendation in Teams, confirms GL account lock, approves unlock action.

5

5. Resolve

Approved action executes: GL account 4100 unlocked, IDoc reposted successfully.

6

6. Validate

System confirms: IDoc posted to ERP, order now visible in sales order list, no downstream errors detected.

AGENT DESIGN

Specialized Agents for End-to-End Incident Resolution



Triage & Classification

Understands alert type, service, urgency, and impact.



Context & Correlation

Collects alert details, logs, events, and change context.



Knowledge & Root Cause

Retrieves runbooks, known errors, and hypotheses.



Resolution & Coordination

Recommends actions, identifies owners, communicates.



Remediation & Validation

Executes approved actions, verifies outcomes.

GROUNDING INTELLIGENCE

Evidence-Based Recommendations, Not Generic Answers

Operational Context

- Cloud ALM alert and event details
- Integration and application logs

Governed AI and Retrieval

- SAP AI Core and Generative AI Hub for governed model access
- HANA Cloud vector engine for semantic retrieval

OPERATOR EXPERIENCE

Contextual Incident Resolution in Microsoft Teams

Instead of a raw alert, the operator receives a decision-ready incident briefing.

Alert Details

Affected Scope

Evidence Summary

Example Prompts

"Show related incidents"

"Explain the likely cause"

"Prepare the approved remediation plan"

END-TO-END FLOW

From Qualifying Alert to Verified Resolution



GOVERNANCE

Human Approval Is an Architectural Control

Grounded recommendations

Uses authorized data and curated knowledge.

Policy-based decisions

Policies define eligibility, routing, and approvals.

Allow-listed remediation

Invokes defined parameterized workflows.

Approval before consequential action

Authorized users approve production changes.

Full audit trail

Captures alert, evidence, approvals, and outcomes.

IMPLEMENTATION

Start Narrow, Prove Value, Then Scale



Phase 1 — Monitor and notify

Select high-value alert categories; trigger Teams notifications and ServiceNow updates.



Phase 2 — Investigate and recommend

Add triage, correlation, evidence gathering, retrieval, and grounded recommendations.



Phase 3 — Governed resolution

Introduce approval workflows and low-risk, allow-listed diagnostic or recovery actions.



Phase 4 — Learn and scale

Capture outcomes, improve knowledge, and expand to additional alert categories.

MEASUREMENT

Measure Operational Value and Trust

Incident Efficiency

- Time to acknowledge
- Time to resolution

Resolution Quality

- Recommendation acceptance rate
- Correct routing rate

Governance and Adoption

Approval Compliance

Tracking adherence to required approval gates before consequential actions.

Audit Completeness

Ensuring every incident has a full traceable record from alert to closure.

KEY TAKEAWAYS

Build AI Agents That Help Operators Resolve Incidents — Safely

1 Cloud ALM monitors; agents investigate and recommend

Use qualifying alerts to invoke targeted investigation and resolution workflows.

2 Design for multiple alert types

The pattern applies to integration, application, job, health, connectivity, and business-process issues.

3 Follow the six-step lifecycle

Monitor, Investigate, Recommend, Approve, Resolve, Validate ensures structured incident handling.

4 Ground recommendations in evidence

Use approved runbooks, historical outcomes, and authorized operational data.

5 Keep humans accountable

Use approval gates, allow-listed remediation, validation, and full auditability.

CLOSING

Turn Operational Alerts Into Contextual, Human-Governed Action

The purpose is not autonomous change for its own sake. It is faster, more consistent, evidence-based incident resolution for SAP operations.



Monitor

Identify new and evolving qualifying operational issues.



Investigate

Assemble evidence, knowledge, and likely resolution paths.



Collaborate

Bring the right context to the right people in Teams and ServiceNow.



Resolve

Approve, execute, validate, document, and learn from each incident.

Thank you – Questions and discussion